Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (2 pts)**.** Let G and H be graphs. A *graph homomorphism* from G to H is a function $f\colon V(G) \to V(H)$ such that $\{f(u), f(v)\} \in E(H)$ whenever $\{u, v\} \in E(G)$. Prove that a graph G is bipartite if and only if there is a graph homomorphism from G to K_2.

(Note that a graph homomorphism does *not* need to be a bijection and that it could be the case that $\{f(u), f(v)\} \in E(H)$ even though $\{u, v\} \notin E(G)$)

Solution. Let $V(K_2) = \{x_1, x_2\}$ denote the vertex-set for the clique K_2.

($\Rightarrow$) Assume G is bipartite with partition sets A_1 and A_2. Define a function $f\colon V(G) \to V(K_2)$ by

$$f(v) = \begin{cases} x_1 \text{ if } v \in A_1, \\ x_2 \text{ if } v \in A_2. \end{cases}$$

We show that f is a graph homomorphism. Indeed, if $uv \in E(G)$, since $V(G) = A_1 \sqcup A_2$ is a bipartition, u and v belong to different parts; we might assume $u \in A_1$ and $v \in A_2$. Thus $f(u) = x_1$ and $f(v) = x_2$, so $f(u)f(v) = x_1x_2 \in E(K_2)$ as desired.

($\Leftarrow$) Assume $f\colon V(G) \to V(K_2)$ defines a graph homomorphism. For $i = 1, 2$, consider

$$A_i = \{u \in V(G)\colon f(u) = x_i\}.$$

Let us show that G is bipartite with parts A_1 and A_2. Since f maps every vertex in G to a unique vertex in K_2, we have $V(G) = A_1 \sqcup A_2$. Thus it only remains to show that A_1 and A_2 are independent. Given $uv \in E(G)$, since f is a homomorphism we have $f(u)f(v) \in E(K_2)$. Thus $f(u) \neq f(v)$, which means that u and v belong to different A_i's. □

Problem 2 (2 pts)**.** A graph G is called *self-complementary* if $G \cong \overline{G}$. For example, P_4 and C_5 are self-complementary.

Prove that if G is a self-complementary graph on n vertices, then n is congruent to either 0 or 1 modulo 4.

Solution. Note first that for any graph G on n vertices we have $E(\overline{G}) = E(K_n) - E(G)$, thus $|E(\overline{G})| = \binom{n}{2} - |E(G)|$.

Now if $G \cong \overline{G}$, we have in particular $|E(G)| = |E(\overline{G})|$. Thus

$$|E(G)| = |E(\overline{G})| = \binom{n}{2} - |E(G)| \qquad \Rightarrow \qquad |E(G)| = \frac{1}{2}\binom{n}{2} = \frac{n(n-1)}{4}.$$

Since $|E(G)|$ is an integer, this means that 4 divides $n(n-1)$. Observe that only of n or $n-1$ is even, and the other one is odd. So the only way for $4 \mid n(n-1)$ is if $4 \mid n$ or $4 \mid n-1$. The former means that $n \equiv 0 \pmod 4$ and the latter $n \equiv 1 \pmod 4$. □

Problem 3 (2 pts)**.** Let T be a tree on n vertices with $\Delta(T) \leqslant 2$. Prove that $T \cong P_n$.

Solution. We proceed by induction on n.

The base case $n = 1$ follows directly, since the unique tree in one vertex is (isomporphic to) P_1.

Assume next that T is a tree on $n \geqslant 2$ vertices with $\Delta(T) \leqslant 2$. We know from class that T contains a leaf ℓ, and that the graph $T' = T - \ell$ is a tree on $n-1$ vertices. Moreover, $\Delta(T') \leqslant \Delta(T) \leqslant 2$. Hence, by inductive hypothesis, we have $T' \cong P_{n-1}$. Thus we can label $V(T') = \{v_1, \ldots, v_{n-1}\}$ such that $E(T') = \{v_iv_{i+1}\colon 1 \leqslant i \leqslant n-2\}$.

Now let w denote the unique neighbor of ℓ in T. Since w is actually a vertex in T', we have $w = v_j$ for some $j \in \{1, \dots, n-1\}$. Now we observe that j must be outside of $\{2, \dots, n-2\}$. Indeed, otherwise, we have $v_{j-1}, v_{j+1}, \ell \in N_T(w)$ which implies $\deg_T w \geqslant 3$ since these three vertices are different. But this contradicts $\Delta(T) \leqslant 2$.

Thus $j \in \{1, n-1\}$. If $j = 1$ then T is the path $(\ell, w = v_1, v_2, \dots, v_n - 1)$ and if $j = n$ then T is the path $(v_1, v_2, \dots, v_{n-2}, v_{n-1} = w, \ell)$, since the only neighbor of ℓ in T is $w = v_j$. In any case, $T \cong P_n$, which concludes the proof. □

Problem 4 (2 pts)**.** Determine (with proof) all trees T (up to isomorphism) on $n \geqslant 2$ vertices such that $\overline{T}$ is also a tree. (Note: we do not require that $T \cong \overline{T}$.)

Solution. Assume that T is a tree on n vertices such that $\overline{T}$ is also a tree. Observe that both T and $\overline{T}$ have $n-1$ edges. But then

$$n - 1 = |E(\overline{T})| = \binom{n}{2} - |E(T)| = \binom{n}{2} - (n-1) \quad \Rightarrow \quad 2(n-1) = \frac{n(n-1)}{2} \quad \Rightarrow \quad n = 4,$$

where the last implication uses that $n - 1 \neq 0$. Since T is connected and $|V(T)| = 4$, we know that $1 \leqslant \Delta(T) \leqslant 3$. We consider two possibilities.

$\diamondsuit$ Case 1: $\Delta(T) = 3$. Since T has 4 vertices, and 3 edges, there is a unique vertex with degree 3 and every edge is incident to that vertex of T with degree 3. Thus $T \cong K_{1,3}$. But in this case $\overline{T} \simeq K_1 \sqcup K_3$ is not connected, hence not a tree.

$\diamondsuit$ Case 2: $\Delta(T) \geqslant 2$. Now we can use Problem 3 to deduce that $T \simeq P_4$; since P_4 is self-complementary, in this case $\overline{T}$ is also a tree.

We conclude that, up to isomorphism, the only tree T on at least two vertices with the property that $\overline{T}$ is also a tree is P_4. □

Problem 5 (2 pts)**.** Let G and H be two self-complementary graphs on disjoint vertex sets, where H has even order n. Let F be the graph obtained from $G \cup H$ by joining each vertex of G to every vertex of degree less than $n/2$ in H. Show that F is self-complementary.

Hint: A graph F is self-complementary if there exists a $\varphi : V(F) \to V(F)$ such that $vw \in E(F)$ if and only if $\varphi(v)\varphi(w) \notin E(F)$.

Solution. Let φ_G be a map that shows G is self-complementary. That is vw is an edge in G if and only if $\varphi_G(v)\varphi_G(w)$ is a nonedge in G. Similarly, let φ_H be a map that shows H is self-complementary.

Now, consider the following map on F:

$$\varphi(v) = \begin{cases} \varphi_G(v) \text{ if } v \in V(G), \\ \varphi_H(v) \text{ if } v \in V(H). \end{cases}$$

Let us show that $vw \in E(F)$ if and only if $\varphi(v)\varphi(w) \notin E(F)$. Consider $v, w \in V(F)$. Note that as long as $v, w \in V(G)$ or $v, w \in V(H)$, then $vw \in E(F)$ if and only if $\varphi(v)\varphi(w) \notin E(F)$. The only case remaining is if $v \in V(G)$ and $w \in V(H)$.

If $vw \in E(F)$, then $\deg(w) \leqslant \frac{n}{2} - 1$ in H. Since $\varphi_H(w)$ has degree $n - 1 - \deg(w) \geqslant \frac{n}{2}$ in H, we have $\varphi(v)\varphi(w) \notin E(F)$. If $vw \notin E(F)$, then $\deg(w) \geqslant n/2$ in H. Since $\varphi_H(w)$ has degree $n - 1 - \deg(w) \leqslant \frac{n}{2} - 1$ in H, we have $\varphi(v)\varphi(w) \in E(F)$.

Thus φ shows that F is self-complementary. □